

Alapan Das

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Employement

Laboratoire de Physique de l'École Normale Supérieure, ENS-PSL, Paris,
Doctorat en physique

Sept 2025 – present

Thesis title: Unusual phase transitions due to quantum measurements
Advisor: Prof. Adam Nahum

Education

Tata Institute of Fundamental Research, Mumbai
Master of Science in Theoretical Physics

Aug 2022 – July 2025

- GPA: 8.4/10

Thesis title: Ising model on hyperbolic lattices (Grade: 9/10)
Advisor: Prof. Kedar Damle.

Visva Bharati University, Santiniketan
Bachelor of Science, Physics

July 2019 – Aug 2022

- GPA: 8.55/10

Birbhum Zilla School, Suri
10+2, Science

2009 – 2019

- Grade: 94%

Publications and pre-prints

Bayesian phase transition for the critical Ising model: Enlarged replica symmetry in the epsilon expansion and in 2D, Kay Jörg Wiese, AD, Adam Nahum ; [arXiv:2604.23346](#) (to be updated with version 2)

Projects

Ising Model on Hyperbolic Lattices (Master's thesis)

Jan 2025-June 2025

- Studied the free fermionic spectrum of the Ising model on the hyperbolic lattices. In this project, I constructed a dimer mapping for the hyperbolic lattice that respects the Fuchsian group symmetry. I employed the generalised Bloch theorem for non-Abelian Fuchsian groups to study the free fermionic spectrum of the system to understand several properties of this system. (*Manuscript under preparation, with Prof. Kedar Damle*)

Free Fermionic Representation of RBIM

Feb 2024-Dec 2024

- We constructed a dimer mapping with a real Hamiltonian for random bond Ising models in two dimensions. From this, we studied the low-energy modes of this free fermionic model, which are introduced by frustration. We connected the frustration to a monomer travelling problem and found a lower bound for the cumulative distribution function of these modes, which helps us understand Griffith's phase better. (*Manuscript under preparation, with Prof. Kedar Damle*)

Academic achievements

Joint Entrance Screening Test, Physics
Score-74/100, All India Rank - 1

Mar 2022

IIT-IISc Joint Admission test for Master's, Physics
Score-89.67/100, All India Rank - 2

Feb 2022

Jagadish Bose National Science Talent Search (JBNSTS), Kolkata
Senior and Junior Scholar

2019 & 2017

DST-Inspire scholarship
Govt. of India

2019-2022

Conferences and Workshops

School on Frustrated Quantum Magnets
École de Physique des Houches, Les Houches, France

Oct 28 - Nov 8, 2024

Stability of Quantum Matters In and Out of Equilibrium at Various Scales
ICTS-TIFR, Bangalore

15-24 Jan, 2024

IISER-JBNSTS workshop, IISER Kolkata
Workshop organised for JBNSTS junior scholars

July 2018

Work Experience

Teaching assistant for Advanced Quantum Mechanics course
Instructor: Prof. Rajdeep Sensarma

Aug-Dec 2024

Teaching assistant for Mathematical methods course
Instructor: Prof. Shravan Hanasoge

Aug-Dec 2023

Skills

C++, Python, Mathematica, Inkscape

Extracurricular activities

Secretary of TIFR movie club
I was a co-secretary of the movie club of the TIFR student society

Aug 2024 - May 2025